

EXT 1: Statistical Analysis (Ext1), S1 The Binomial (Y12) Binomial Probability (Ext1)

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Exam Equivalent Time: 102 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

HISTORICAL CONTRIBUTION

- *Statistical Analysis* is a new syllabus category that incorporates some old *Ext1* content on binomial probability with new content looking at statistical analysis of Bernoulli random variables and normal approximations of sample proportions.
- This topic has been split into three categories for analysis purposes which are: 1- *Binomial Probability*, 2-*Statistics and Binomial Distributions* and 3-*Normal Approximations of Sample Proportions*.
- This analysis looks at *Binomial Probability*.

HSC ANALYSIS - What to expect and common pitfalls

- *Binomial Probability* is old course Ext1 probability content that models two outcome situations.
- This topic was not examined in the 2020 Ext1 exam, although historically it had been examined virtually every year in the old course, accounting for a substantial average of 3.8% per year. This statistic comes with the caveat that *Binomial Probability* will have to compete with the other major Statistics topic areas for mark allocations going forward.
- The results of past questions have been volatile - we note that sub-50% mean marks have resulted around half the time.
- A number of harder examples require students to express probabilities in algebraic terms and prove identities. We recommend carefully reviewing *2006 Ext1 6b*, and *2011 Ext1 6c* to cover this difficult area.

Questions

1. Statistics, EXT1 S1 2021 HSC 6 MC

The random variable X represents the number of successes in 10 independent Bernoulli trials. The probability of success is $p = 0.9$ in each trial.

Let $r = P(X \geq 1)$.

Which of the following describes the value of r ?

- A. $r > 0.9$
- B. $r = 0.9$
- C. $0.1 < r < 0.9$
- D. $r \leq 0.1$

2. Statistics, EXT1 S1 MET2 2008 14 MC

The minimum number of times that a fair coin can be tossed so that the probability of obtaining a head on each trial is less than 0.0005 is

- A. 9
- B. 10
- C. 11
- D. 12

3. Statistics, EXT1 S1 2013 HSC 11c

An examination has 10 multiple-choice questions, each with 4 options. In each question, only one option is correct. For each question a student chooses one option at random.

Write an expression for the probability that the student chooses the correct option for exactly 7 questions. (2 marks)

4. Statistics, EXT1 S1 2017 HSC 11g

The probability that a particular type of seedling produces red flowers is $\frac{1}{5}$.

Eight of these seedlings are planted.

- i. Write an expression for the probability that exactly three of the eight seedlings produce red flowers. (1 mark)
- ii. Write an expression for the probability that none of the eight seedlings produces red flowers. (1 mark)
- iii. Write an expression for the probability that none of the eight seedlings produces red flowers. (1 mark)

5. Statistics, EXT1 S1 2016 HSC 11f

A darts player calculates that when she aims for the bullseye the probability of her hitting the bullseye is $\frac{3}{5}$ with each throw.

- Find the probability that she hits the bullseye with exactly one of her first three throws. *(1 mark)*
 - Find the probability that she hits the bullseye with at least two of her first six throws. *(2 marks)*
-

6. Statistics, EXT1 S1 2019 HSC 11f

Prize-winning symbols are printed on 5% of ice-cream sticks. The ice-creams are randomly packed into boxes of 8.

- What is the probability that a box contains no prize-winning symbols? *(1 mark)*
 - What is the probability that a box contains at least 2 prize-winning symbols? *(2 marks)*
-

7. Statistics, EXT1 S1 2010 HSC 1f

Five ordinary six-sided dice are thrown.

What is the probability that exactly two of the dice land showing a four?

Leave your answer in unsimplified form. *(1 mark)*

8. Statistics, EXT1 S1 2014 HSC 11b

The probability that it rains on any particular day during the 30 days of November is 0.1.

Write an expression for the probability that it rains on fewer than 3 days in November. *(2 marks)*

9. Statistics, EXT1 S1 2018 HSC 12d

A group of 12 people sets off on a trek. The probability that a person finishes the trek within 8 hours is 0.75.

Find an expression for the probability that at least 10 people from the group complete the trek within 8 hours. *(2 marks)*

10. Statistics, EXT1 S1 SM-Bank 3

In a chocolate factory the material for making each chocolate is sent to a machines.

The time, X seconds, taken to produce a chocolate by machine is a binomial distribution where it can be shown that $P(X \leq 3) = \frac{9}{32}$.

A random sample of 10 chocolates is chosen. Find the probability, correct to two decimal places, that exactly 4 of these 10 chocolates took 3 or less seconds to produce. *(2 marks)*

11. Statistics, EXT1 S1 2020 MET1 5

For a certain population the probability of a person being born with the specific gene SPGE1 is $\frac{3}{5}$.

The probability of a person having this gene is independent of any other person in the population having this gene.

In a randomly selected group of four people, what is the probability that three or more people have the SPGE1 gene? *(2 marks)*

12. Statistics, EXT1 S1 EQ-Bank 12

A manufacturer of pool cue tips knows that 4% of pool cue tips produced in its factory need to be scrapped.

A random sample of 10 cue tips produced in the factory is examined.

- What is the probability that exactly 3 cue tips need to be scrapped?
Give your answer correct to 3 decimal places. *(1 mark)*
 - What is the probability that, at most, 2 cue tips need to be scrapped?
Give your answer correct to 3 decimal places. *(2 marks)*
-

13. Statistics, EXT1 S1 SM-Bank 2

A school has a class set of 22 new laptops kept in a recharging trolley. Provided each laptop is correctly plugged into the trolley after use, its battery recharges.

On a particular day, a class of 22 students uses the laptops. All laptop batteries are fully charged at the start of the lesson. Each student uses and returns exactly one laptop. The probability that a student does **not** correctly plug their laptop into the trolley at the end of the lesson is 10%. The correctness of any student's plugging-in is independent of any other student's correctness.

Determine the probability that at least one of the laptops is **not** correctly plugged into the trolley at the end of the lesson. Give your answer correct to three decimal places. *(2 marks)*

14. Statistics, EXT1 S1 2007 MET1 5

It is known that 50% of the customers who enter a restaurant order a cup of coffee. If four customers enter the restaurant, what is the probability that more than two of these customers order coffee? (Assume that what any customer orders is independent of what any other customer orders.) *(2 marks)*

15. Statistics, EXT1 S1 2016 MET1 4

A paddock contains 10 tagged sheep and 20 untagged sheep. Four times each day, one sheep is selected at random from the paddock, placed in an observation area and studied, and then returned to the paddock.

- What is the probability that the number of tagged sheep selected on a given day is zero? (1 mark)
- What is the probability that at least one tagged sheep is selected on a given day? (1 mark)
- What is the probability that no tagged sheep are selected on each of six consecutive days?

Express your answer in the form $\left(\frac{a}{c}\right)^c$, where a, b and c are positive integers. (1 mark)

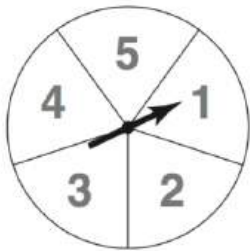
16. Statistics, EXT1 S1 2007 HSC 4a

In a large city, 10% of the population has green eyes.

- What is the probability that two randomly chosen people both have green eyes? (1 mark)
- What is the probability that exactly two of a group of 20 randomly chosen people have green eyes? Give your answer correct to three decimal places. (1 mark)
- What is the probability that more than two of a group of 20 randomly chosen people have green eyes? Give your answer correct to two decimal places. (2 marks)

17. Statistics, EXT1 S1 2012 HSC 12c

Kim and Mel play a simple game using a spinner marked with the numbers 1, 2, 3, 4 and 5.



The game consists of each player spinning the spinner once. Each of the five numbers is equally likely to occur.

The player who obtains the higher number wins the game.

If both players obtain the same number, the result is a draw.

- Kim and Mel play one game. What is the probability that Kim wins the game? (1 mark)
- Kim and Mel play six games. What is the probability that Kim wins exactly three games? (2 marks)

18. Statistics, EXT1 S1 2005 HSC 6a

There are five matches on each weekend of a football season. Megan takes part in a competition in which she earns one point if she picks more than half of the winning teams for a weekend, and zero points otherwise. The probability that Megan correctly picks the team that wins any given match is $\frac{2}{3}$.

- Show that the probability that Megan earns one point for a given weekend is 0.7901, correct to four decimal places. (2 marks)
- Hence find the probability that Megan earns one point every week of the eighteen-week season. Give your answer correct to two decimal places. (1 mark)
- Find the probability that Megan earns at most 16 points during the eighteen-week season. Give your answer correct to two decimal places. (2 marks)

19. Statistics, EXT1 S1 2006 HSC 6b

In an endurance event, the probability that a competitor will complete the course is p and the probability that a competitor will not complete the course is $q = 1 - p$. Teams consist of either two or four competitors. A team scores points if at least half its members complete the course.

- Show that the probability that a four-member team will have at least three of its members not complete the course is $4pq^3 + q^4$. (1 mark)
- Hence, or otherwise, find an expression in terms of q only for the probability that a four-member team will score points. (2 marks)
- Find an expression in terms of q only for the probability that a two-member team will score points. (1 mark)
- Hence, or otherwise, find the range of values of q for which a two-member team is more likely than a four-member team to score points. (2 marks)

20. Statistics, EXT1 S1 2009 HSC 4a

A test consists of five multiple-choice questions. Each question has four alternative answers. For each question only one of the alternative answers is correct.

Huong randomly selects an answer to each of the five questions.

- What is the probability that Huong selects three correct and two incorrect answers? (2 marks)
- What is the probability that Huong selects three or more correct answers? (2 marks)
- What is the probability that Huong selects at least one incorrect answer? (1 mark)

21. Statistics, EXT1 S1 2011 MET1 7

A biased coin tossed three times. The probability of a head from a toss of this coin is p .

- Find, in terms of p , the probability of obtaining
 - three heads from the three tosses (1 mark)
 - two heads and a tail from the three tosses. (1 mark)
- If the probability of obtaining three heads equals the probability of obtaining two heads and a tail, find p . (2 marks)

22. Statistics, EXT1 S1 SM-Bank 1

Shoddy Ltd produces statues that are classified as Superior or Regular and are entirely made by machines, on a construction line. The quality of any one of Shoddy's statues is independent of the quality of any of the others on its construction line. The probability that any one of Shoddy's statues is Regular is 0.8.

Shoddy Ltd wants to ensure that the probability that it produces at least one Superior statues in a day's production run is at least 0.95.

Calculate the minimum number of statues that Shoddy would need to produce in a day to achieve this aim. (3 marks)

23. Statistics, EXT1 S1 2011 HSC 6c

A game is played by throwing darts at a target. A player can choose to throw two or three darts.

Darcy plays two games. In Game 1, he chooses to throw two darts, and wins if he hits the target at least once. In Game 2, he chooses to throw three darts, and wins if he hits the target at least twice.

The probability that Darcy hits the target on any throw is p , where $0 < p < 1$.

- Show that the probability that Darcy wins Game 1 is $2p - p^2$. (1 mark)
- Show that the probability that Darcy wins Game 2 is $3p^2 - 2p^3$. (1 mark)
- Prove that Darcy is more likely to win Game 1 than Game 2. (2 marks)
- Find the value of p for which Darcy is twice as likely to win Game 1 as he is to win Game 2. (2 marks)

Worked Solutions

1. Statistics, EXT1 S1 2021 HSC 6 MC

$$p = 0.9, \quad \bar{p} = 0.1$$

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - (0.1)^{10} \\ &= 0.999\dots \end{aligned}$$

$$\therefore r > 0.9$$

$$\Rightarrow A$$

2. Statistics, EXT1 S1 MET2 2008 14 MC

Let X = Number of heads

$$X \sim \text{Bin}(n, p) \sim \text{Bin}\left(n, \frac{1}{2}\right)$$

$$P(X = n) < 0.0005$$

$$\binom{n}{n} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^0 < 0.0005$$

$$\frac{1}{2^n} < \frac{5}{10\,000}$$

$$2^n > 2000$$

$$\ln 2^n > \ln 2000$$

$$n > \frac{\ln 2000}{\ln 2}$$

$$n > 10.97$$

$$\therefore n_{\min} = 11$$

$$\Rightarrow C$$

3. Statistics, EXT1 S1 2013 HSC 11c

$$P(\text{Correct}) = \frac{1}{4}$$

$$P(\text{Incorrect}) = \frac{3}{4}$$

$$\therefore P(\text{exactly 7 correct}) = {}^{10}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^3$$

4. Statistics, EXT1 S1 2017 HSC 11g

i. $P(\text{Red}) = \frac{1}{5}$, $P(\text{Not Red}) = \frac{4}{5}$

$$P(\text{exactly 3 are red})$$

$$= {}^8C_3 \cdot \left(\frac{1}{5}\right)^3 \cdot \left(\frac{4}{5}\right)^5$$

ii. $P(\text{none are red})$

$$= {}^8C_0 \cdot \left(\frac{1}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^8$$

$$= \left(\frac{4}{5}\right)^8$$

iii. $P(\text{at least 1 is red})$

$$= 1 - P(\text{none are red})$$

$$= 1 - \left(\frac{4}{5}\right)^8$$

5. Statistics, EXT1 S1 2016 HSC 11f

i. $P(\text{exactly 1 bullseye})$

$$= {}^3C_1 \cdot \left(\frac{3}{5}\right)^1 \left(\frac{2}{5}\right)^2$$

$$= 3 \cdot \left(\frac{3}{5}\right) \cdot \left(\frac{4}{25}\right)$$

$$= \frac{36}{125}$$

ii. $P(\text{at least 2 from 6 throws})$

$$= 1 - [P(0) + P(1)]$$

$$= 1 - \left[\left(\frac{2}{5}\right)^6 + {}^6C_1 \cdot \left(\frac{3}{5}\right)^1 \cdot \left(\frac{2}{5}\right)^5 \right]$$

$$= 1 - \left[\frac{128}{3125} \right]$$

$$= \frac{2997}{3125}$$

6. Statistics, EXT1 S1 2019 HSC 11f

i. Chances of any stick winning:

$$P(W) = 0.05$$

$$P(\overline{W}) = 0.95$$

$$P(\text{In box of 8, all } \overline{W})$$

$$= 0.95^8$$

ii. $P(\text{at least two winners in a box})$

$$= 1 - P(1 \text{ winner}) - P(0 \text{ winners})$$

$$= 1 - {}^8C_1 \times 0.95^7 \times 0.05^1 - {}^8C_0 \times 0.95^8$$

$$= 0.05724 \dots$$

$$= 5.72\% \text{ (to 2 d.p.)}$$

7. Statistics, EXT1 S1 2010 HSC 1f

$$P(4) = \frac{1}{6}$$

$$P(\overline{4}) = \frac{5}{6}$$

Combinations of two 4's = 5C_2

$$\therefore P(\text{exactly two 4s}) = {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

COMMENT: Surprisingly, half of students sitting the exam got this wrong. Easily the most poorly answered part of Q1 in 2010.

8. Statistics, EXT1 S1 2014 HSC 11b

$$P(R) = 0.1$$

$$P(\overline{R}) = 1 - 0.1 = 0.9$$

Over 30 days:

$$\begin{aligned} P(R < 3) &= P(R = 2) + P(R = 1) + P(R = 0) \\ &= {}^{30}C_2(0.1)^2(0.9)^{28} + {}^{30}C_1(0.1)^1(0.9)^{29} \\ &\quad + {}^{30}C_0(0.1)^0(0.9)^{30} \\ &= {}^{30}C_2(0.1)^2(0.9)^{28} + {}^{30}C_1(0.1)(0.9)^{29} + (0.9)^{30} \end{aligned}$$

9. Statistics, EXT1 S1 2018 HSC 12d

$$P(\text{finishes within 8 hours}) = 0.75$$

$$P(\text{not finish within 8 hours}) = 0.25$$

Let X = number who finish below 8 hours

$$\begin{aligned} \therefore P(\text{at least 10 finish within 8 hours}) &= P(X=10) + P(X=11) + P(X=12) \\ &= \binom{12}{10}(0.75)^{10}(0.25)^2 + \binom{12}{11}(0.75)^{11}(0.25)^1 + \binom{12}{12}(0.75)^{12} \end{aligned}$$

10. Statistics, EXT1 S1 SM-Bank 3

Let C = number of chocolates that take less than 3 seconds

$$C \sim \text{Bin}(n, p) \sim \text{Bin}\left(10, \frac{9}{32}\right)$$

$$\begin{aligned} P(C = 4) &= \binom{10}{4} \left(\frac{9}{32}\right)^4 \left(\frac{23}{32}\right)^6 \\ &= 0.181... \\ &= 0.18 \quad (2 \text{ d.p.}) \end{aligned}$$

COMMENT: Take care as $P(X \leq 3) = \frac{9}{32}$ provides the equivalent of p here.

11. Statistics, EXT1 S1 2020 MET1 5

Let X = number of people with gene

$$X \sim \text{Bin}\left(4, \frac{3}{5}\right)$$

$$\begin{aligned} P(X \geq 3) &= P(X = 3) + P(X = 4) \\ &= {}^4C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right) + {}^4C_4 \left(\frac{3}{5}\right)^4 \\ &= \frac{4 \times 27 \times 2}{625} + \frac{81}{625} \\ &= \frac{297}{625} \end{aligned}$$

12. Statistics, EXT1 S1 EQ-Bank 12

i. Let X = number of defective cue tips

$$X \sim \text{Bi}(10, 0.04)$$

$$\begin{aligned} P(X = 3) &= {}^{10}C_3 \cdot (0.04)^3 (0.96)^7 \\ &= 0.00577... \\ &= 0.006 \quad (\text{to 3 d.p.}) \end{aligned}$$

ii. $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$$\begin{aligned} &= {}^{10}C_0(0.04)^0(0.96)^{10} + {}^{10}C_1(0.04)(0.96)^9 + {}^{10}C_2(0.04)^2(0.96)^8 \\ &= 0.66483 + 0.27701 + 0.05194 \\ &= 0.994 \quad (\text{to 3 d.p.}) \end{aligned}$$

13. Statistics, EXT1 S1 SM-Bank 2

Let X = number not correctly plugged

$$X \sim \text{Bin}(n, p) \sim \text{Bin}(22, 0.1)$$

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - \binom{22}{0} (0.1^0) (0.9^{22}) \\ &= 1 - 0.9^{22} \\ &= 0.9015... \\ &= 0.902 \text{ (3 d.p.)} \end{aligned}$$

14. Statistics, EXT1 S1 2007 MET1 5

$$X \sim \text{Bin}(n, p) \sim \text{Bin}\left(4, \frac{1}{2}\right)$$

$$\begin{aligned} P(X > 2) &= P(X = 3) + P(X = 4) \\ &= \binom{4}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right) + \binom{4}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 \\ &= 4 \times \frac{1}{16} + 1 \times \frac{1}{16} \\ &= \frac{5}{16} \end{aligned}$$

15. Statistics, EXT1 S1 2016 MET1 4

a. Let X = Number of tagged sheep,

$$X \sim \text{Bin}(n, p) \sim \text{Bin}\left(4, \frac{1}{3}\right)$$

$$\begin{aligned} P(X = 0) &= \binom{4}{0} \times \left(\frac{1}{3}\right)^0 \times \left(\frac{2}{3}\right)^4 \\ &= \frac{16}{81} \end{aligned}$$

$$\begin{aligned} \text{b. } P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - \frac{16}{81} \\ &= \frac{65}{81} \end{aligned}$$

c. Let Y = Days that no tagged sheep selected,

$$Y \sim \text{Bin}\left(6, \frac{16}{81}\right)$$

$$\begin{aligned} P(Y = 6) &= \binom{6}{6} \times \left(\frac{16}{81}\right)^6 \times \left(\frac{65}{81}\right)^0 \\ &= \left(\frac{16}{81}\right)^6 \\ &= \left(\frac{2}{3}\right)^{24} \end{aligned}$$

16. Statistics, EXT1 S1 2007 HSC 4a

i. $P(G) = 0.1$

$$\begin{aligned} P(GG) &= 0.1 \times 0.1 \\ &= 0.01 \end{aligned}$$

ii. $P(\text{not } G) = 1 - 0.1 = 0.9$

$$\begin{aligned} \therefore P(2 \text{ out of } 20 \text{ have green eyes}) \\ &= {}^{20}C_2 \cdot (0.1)^2 \cdot (0.9)^{18} \\ &= 0.2851\dots \\ &= 0.285 \text{ (to 3 d.p.)} \end{aligned}$$

iii. $P(\text{more than 2 have green eyes})$

$$\begin{aligned} &= 1 - [P(0) + P(1) + P(2)] \\ &= 1 - [0.9^{20} + {}^{20}C_1(0.1)^1(0.9)^{19} + {}^{20}C_2(0.1)^2(0.9)^{18}] \\ &= 1 - [0.1215\dots + 0.2701\dots + 0.2851\dots] \\ &= 1 - 0.6769\dots \\ &= 0.3230 \\ &= 0.32 \text{ (to 2 d.p.)} \end{aligned}$$

17. Statistics, EXT1 S1 2012 HSC 12c

i. Method 1

$$\begin{aligned} P(\text{Kim wins game}) \\ &= P(K \text{ spins } 5) \times P(M < 5) + P(K \text{ spins } 4) \times P(M < 4) + \dots \\ &= \left(\frac{1}{5} \times \frac{4}{5}\right) + \left(\frac{1}{5} \times \frac{3}{5}\right) + \left(\frac{1}{5} \times \frac{2}{5}\right) + \left(\frac{1}{5} \times \frac{1}{5}\right) \\ &= \frac{10}{25} \\ &= \frac{2}{5} \end{aligned}$$

Method 2

$$P(\text{Draw}) = \frac{1}{5}$$

$$\therefore P(\text{Not a draw}) = 1 - \frac{1}{5} = \frac{4}{5}$$

Since Kim and Mel have equal chance

$$\begin{aligned} P(K \text{ wins}) &= \frac{1}{2} \times \frac{4}{5} \\ &= \frac{2}{5} \end{aligned}$$

ii. $P(\text{Kim wins}) = \frac{2}{5}$

$$P(\text{Kim doesn't win}) = \frac{3}{5}$$

After 6 games,

$$\begin{aligned} P(\text{Kim wins exactly 3}) \\ &= {}^6C_3 \left(\frac{2}{5}\right)^3 \left(\frac{3}{5}\right)^3 \\ &= \frac{6!}{3!3!} \times \frac{8}{125} \times \frac{27}{125} \\ &= \frac{864}{3125} \end{aligned}$$

♦ Mean mark 37%

18. Statistics, EXT1 S1 2005 HSC 6a

i. P (earns a point)

$$\begin{aligned}
 &= P(\text{picks 3, 4 or 5 winners}) \\
 &= {}^5C_3 \left(\frac{2}{3}\right)^3 \cdot \left(\frac{1}{3}\right)^2 + {}^5C_4 \left(\frac{2}{3}\right)^4 \cdot \left(\frac{1}{3}\right)^1 \\
 &\quad + {}^5C_5 \left(\frac{2}{3}\right)^5 \cdot \left(\frac{1}{3}\right)^0 \\
 &= 10 \cdot \frac{8}{27} \cdot \frac{1}{9} + 5 \cdot \frac{16}{81} \cdot \frac{1}{3} + 1 \cdot \frac{32}{243} \cdot 1 \\
 &= \frac{80}{243} + \frac{80}{243} + \frac{32}{243} \\
 &= \frac{192}{243} \\
 &= 0.790123\dots \\
 &= 0.7901 \text{ (to 4 d.p.) } \dots \text{ as required.}
 \end{aligned}$$

ii. P (earns a point 18 weeks in a row)

$$\begin{aligned}
 &= (0.7901\dots)^{18} \\
 &= 0.01440\dots \\
 &= 0.01 \text{ (to 2 d.p.)}
 \end{aligned}$$

iii. P (earns at most 16 points)

$$= 1 - P(\text{earns 17 or 18 points})$$

P (earns 17)

$$\begin{aligned}
 &= {}^{18}C_{17} \cdot (0.7901\dots)^{17} \times (1 - 0.7901\dots) \\
 &= 0.0688\dots
 \end{aligned}$$

$$P(\text{earns 18}) = 0.01440\dots \quad (\text{from (ii)})$$

$\therefore P$ (earns at most 16 points)

$$\begin{aligned}
 &= 1 - (0.0688\dots + 0.0144\dots) \\
 &= 0.916\dots \\
 &= 0.92 \text{ (to 2 d.p.)}
 \end{aligned}$$

19. Statistics, EXT1 S1 2006 HSC 6b

i. P (at least 3 don't complete)

$$\begin{aligned}
 &= P(3 \text{ don't complete}) + P(4 \text{ don't complete}) \\
 &= {}^4C_3 q^3 p + {}^4C_4 q^4 \\
 &= 4pq^3 + q^4
 \end{aligned}$$

ii. P (4-member team scores)

$$\begin{aligned}
 &= 1 - P(\text{at least 3 don't complete}) \\
 &= 1 - 4pq^3 + q^4 \\
 &= 1 - [4(1 - q)q^3 + q^4] \\
 &= 1 - (4q^3 - 4q^4 + q^4) \\
 &= 1 - 4q^3 + 3q^4
 \end{aligned}$$

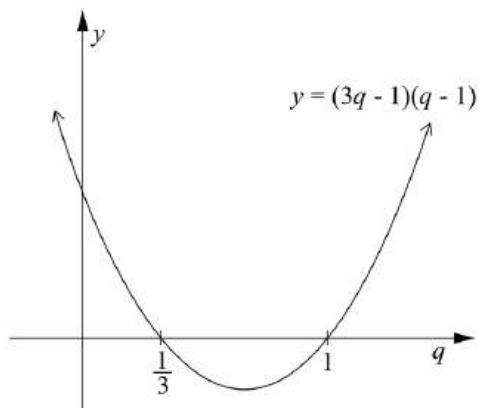
iii. P (2-member team scores)

$$\begin{aligned}
 &= 1 - P(\text{both don't complete}) \\
 &= 1 - q \cdot q \\
 &= 1 - q^2
 \end{aligned}$$

iv. A 2-member team is more likely to score when

$$\begin{aligned}
 1 - q^2 &> 1 - 4q^3 + 3q^4 \\
 3q^4 - 4q^3 + q^2 &< 0 \\
 q^2(3q^2 - 4q + 1) &< 0 \\
 q^2(3q - 1)(q - 1) &< 0
 \end{aligned}$$

$$\text{Consider } (3q - 1)(q - 1) < 0$$



∴ Since q is positive and $\neq 1$

$$q^2(3q - 1)(q - 1) < 0 \text{ when}$$

$$\frac{1}{3} < q < 1.$$

20. Statistics, EXT1 S1 2009 HSC 4a

i. $P(\text{correct}) = \frac{1}{4}$

$$P(\text{wrong}) = \frac{3}{4}$$

$$P(3 \text{ correct, 2 wrong})$$

$$= {}^5C_3 \cdot \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2$$

$$= \frac{5!}{3!2!} \cdot \left(\frac{1}{64}\right) \cdot \left(\frac{9}{16}\right)$$

$$= \frac{90}{1024}$$

$$= \frac{45}{512}$$

ii. $P(3 \text{ or more correct})$

$$= P(3 \text{ correct}) + P(4 \text{ correct}) + P(5 \text{ correct})$$

$$= {}^5C_3 \cdot \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 + {}^5C_4 \cdot \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^1 + {}^5C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^0$$

$$= \frac{90}{1024} + \frac{15}{1024} + \frac{1}{1024}$$

$$= \frac{53}{512}$$

iii. $P(\text{at least 1 incorrect})$

$$= 1 - P(0 \text{ incorrect})$$

$$= 1 - {}^5C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^0$$

$$= 1 - \frac{1}{1024}$$

$$= \frac{1023}{1024}$$

TIP: The use of "at least" should flag a good chance of applying $1 - P(\text{complement})$ to solve.

21. Statistics, EXT1 S1 2011 MET1 7

a.i. $P(HHH) = p \times p \times p$
 $= p^3$

ii. P(2 Heads and 1 Tail from 3 tosses)

$$= \binom{3}{2} \times p^2 \times (1-p)^1$$

$$= 3p^2(1-p)$$

♦ Part (a)(ii) mean mark 41%.

b. If probabilities are equal:

$$p^3 = 3p^2 - 3p^3$$

$$4p^3 - 3p^2 = 0$$

$$p^2(4p - 3) = 0$$

$$\therefore p = 0 \text{ or } p = \frac{3}{4}$$

♦ Mean mark 48%.
MARKER'S COMMENT: Many students incorrectly assumed p could not be zero.

22. Statistics, EXT1 S1 SM-Bank 1

Let X = Number of superior statues

$$X \sim \text{Bin}(n, 0.2)$$

$$P(X \geq 1) \geq 0.95$$

$$1 - P(X = 0) \geq 0.95$$

$$1 - \binom{n}{0}(0.8^n)(0.2^0) \geq 0.95$$

$$1 - 0.8^n \geq 0.95$$

$$0.8^n \leq 0.05$$

$$n \ln 0.8 \leq \ln 0.05$$

$$n \geq \frac{\ln 0.05}{\ln 0.8}, \quad (\ln 0.8 < 0)$$

$$n \geq 13.4...$$

$$\therefore n_{\min} = 14$$

♦♦ Mean mark 27%.

23. Statistics, EXT1 S1 2011 HSC 6c

i. Show $P(\text{wins Game 1}) = 2p - p^2$

$$P(\text{Hits}) = P(H) = p$$

$$P(\text{Miss}) = P(M) = 1 - p$$

$$P(\text{Wins G1}) = 1 - P(\text{MM})$$

$$= 1 - (1 - p)^2$$

$$= 1 - 1 + 2p - p^2$$

$$= 2p - p^2 \quad \dots \text{ as required}$$

ii. Show $P(\text{wins Game 2}) = 3p - 2p^3$:

Darcy wins G2 if he hits 3 times, or twice.

$$P(\text{Hits 3 times}) = {}^3C_3(p)^3 = p^3$$

$$P(\text{Hits twice}) = {}^3C_2(p)^2(1-p) = 3p^2 - 3p^3$$

$$P(\text{Wins G2}) = p^3 + 3p^2 - 3p^3$$

$$= 3p^2 - 2p^3 \quad \dots \text{ as required}$$

iii. Prove more likely to win G1 vs G2

i.e. Show $2p - p^2 > 3p^2 - 2p^3$

$$2p^3 - 4p^2 + 2p > 0$$

$$2p(p^2 - 2p + 1) > 0$$

$$2p(p-1)^2 > 0$$

$$\Rightarrow \text{TRUE since } (p-1)^2 > 0 \text{ and } 0 < p < 1$$

$\therefore$ More likely for Darcy to win Game 1.

iv. If twice as likely to win G1 vs G2

$$2p - p^2 = 2(3p^2 - 2p^3)$$

$$= 6p^2 - 4p^3$$

$$4p^3 - 7p^2 + 2p = 0$$

$$p(4p^2 - 7p + 2) = 0$$

♦ Mean mark 47%.

♦ Mean mark 40%.

♦♦ Mean mark 17%.
COMMENT: It is critical for students to utilise the limits of $0 < p < 1$ to answer part (iii).

♦♦ Mean mark 21%.
MARKER'S COMMENT: BE CAREFUL in formulating your equation here. Many students multiplied the wrong side by 2.

$$\begin{aligned}
 p &= \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \times 4 \times 2}}{2 \times 4} \\
 &= \frac{7 \pm \sqrt{49 - 32}}{8} \\
 &= \frac{7 \pm \sqrt{17}}{8}
 \end{aligned}$$

Since $0 < p < 1$,

$$p = \frac{7 - \sqrt{17}}{8}$$